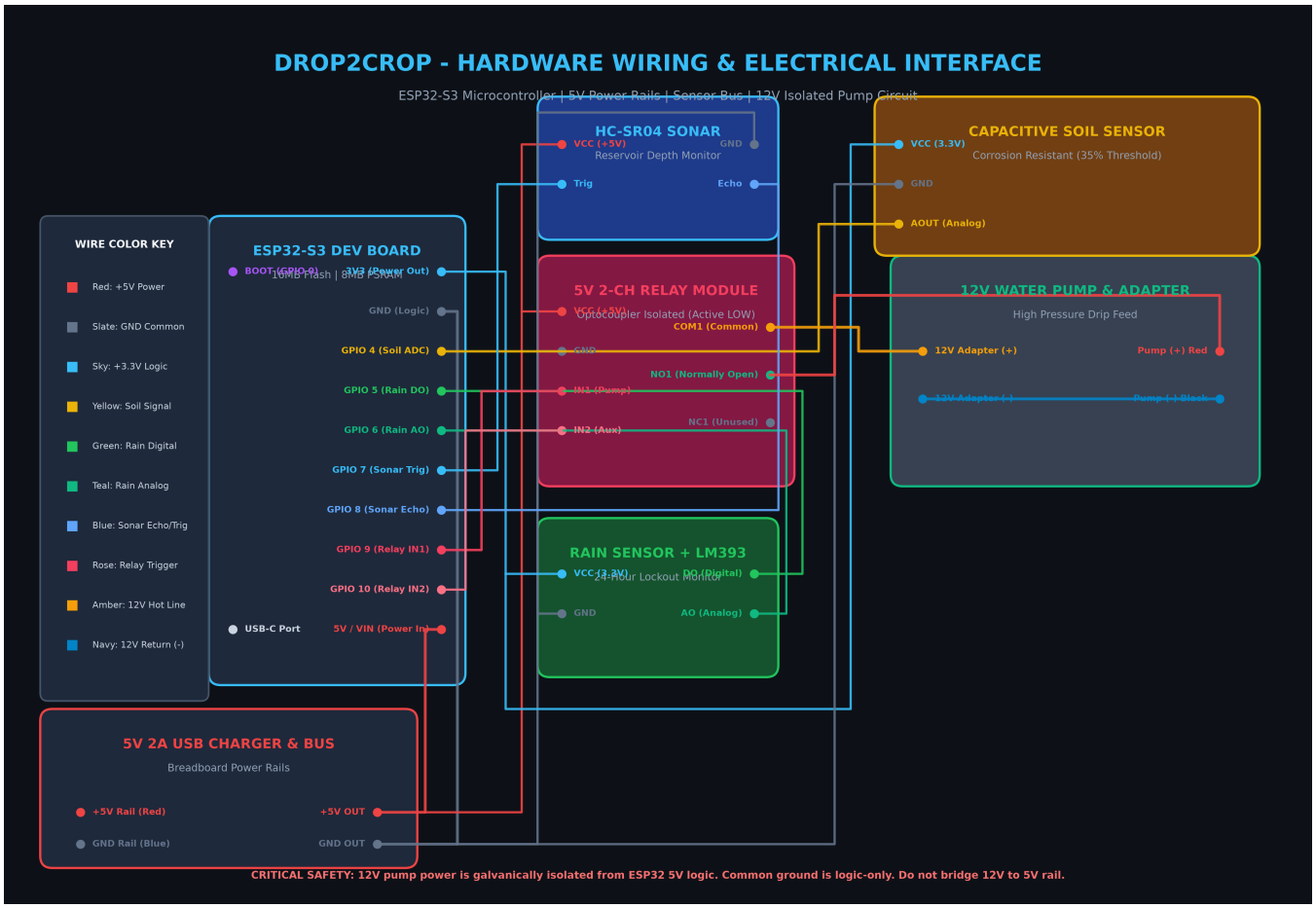
Drop2Crop — Hardware Wiring Diagram & Manual

Complete Electrical Pinout, Power Bus Distribution & Relay Circuitry

1. System Master Wiring Schematic

The diagram below illustrates all physical wire connections between the ESP32-S3, dual-channel relay module, HC-SR04 ultrasonic distance sensor, capacitive soil moisture sensor, rain detector, breadboard power rails, and 12V water pump.



2. Complete ESP32-S3 Pinout Connection Table

Module / Component	Component Pin	ESP32-S3 Pin / Bus	Voltage	Signal Description
Capacitive Soil Sensor	VCC	**ESP32 3V3 Pin**	3.3V DC	Clean 3.3V power (prevents ADC drift)
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	AOUT	**GPIO 4 (ADC1_CH3)**	0 - 3.3V	Analog soil moisture signal (1350-3150)
Rain Sensor (LM393)	VCC	**ESP32 3V3 Pin**	3.3V DC	Logic supply (safe for ESP32 inputs)
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	DO	**GPIO 5**	3.3V Logic	Digital rain trigger (LOW = Rain)
	AO	**GPIO 6 (ADC1_CH5)**	0 - 3.3V	Analog precipitation density
HC-SR04 Ultrasonic	VCC	**Breadboard +5V Rail**	5.0V DC	5V power for sonar transmitter
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	TRIG	**GPIO 7**	3.3V Logic	10µs ultrasonic trigger pulse
	ECHO	**GPIO 8**	Input	Echo pulse return width
5V 2-Channel Relay	VCC	**Breadboard +5V Rail**	5.0V DC	Relay coil power supply
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	IN1	**GPIO 9**	Active LOW	Relay 1 trigger: 12V Water Pump
	IN2	**GPIO 10**	Active LOW	Relay 2 trigger: Aux / Fertigation
Onboard Hardware	BOOT Button	**GPIO 0**	Active LOW	Hold 3s to trigger WiFi Captive Portal

3. Power Distribution Architecture (5V Charger + 12V Pump)

To ensure electrical stability and prevent micro-resets caused by inductive motor surges, the Drop2Crop system separates logic power from actuator power:

- **5V Mobile Phone Charger:** Powers the ESP32-S3 either via its USB-C port or by connecting the charger's +5V and GND wires to the breadboard rails, then linking the rail to the ESP32 **5V / VIN** and **GND** pins.
- **Breadboard Power Rails:** Distribute clean +5V to the Relay Module VCC and HC-SR04 VCC. Provide common Ground to all sensors and modules.
- **3.3V Logic Rail:** The ESP32's onboard low-dropout (LDO) regulator provides 3.3V from its **3V3 pin**. Connect this exclusively to the Capacitive Soil Moisture Sensor and the Rain Sensor comparator board.
- **12V High-Power Pump Isolation:** The 12V Water Pump is powered by an independent 12V Universal Adapter. The 12V (+) wire passes through the Relay 1 **COM (Common)** terminal and exits from the **NO (Normally Open)** terminal to the pump (+) lead. The 12V (-) wire connects directly to the pump (-) lead. **DO NOT connect the 12V power adapter to the breadboard or ESP32 pins.**

ELECTRICAL SAFETY WARNING: Always ensure the 12V adapter is unplugged while inserting jumper wires into the relay screw terminals. The relay module provides optoisolation to protect the ESP32 from back-EMF spikes.